

## 5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 10 of 17

### 5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

#### LEARN IT

**Data protection** means keeping digital information safe from **unauthorised access, misuse, loss, damage or accidental deletion**.

#### 1. Data encryption

**Encryption** changes readable data (**plaintext**) into a coded form called **ciphertext**. Unauthorised users cannot understand ciphertext without the key.

- **Symmetric encryption** — the **same key** is used to encrypt and to decrypt.
- **Asymmetric encryption** — **two keys**: a **public key** to encrypt, a **private key** to decrypt.

A simple exam cipher is the **Caesar cipher** (a symmetric substitution cipher). You shift every letter a fixed number of places along the alphabet. Shift 3 is the classic key: A → D, B → E, ..., Z → C (the alphabet wraps around).

#### 2. Strong passwords

A strong password includes **uppercase, lowercase, numbers and symbols**. It resists **brute-force** guessing (trying every possible combination). Source example: Zk@2026#Pass .

Weak: ahmed123 , pakistan , sindh , your date of birth. Never reuse the same password on the college LMS and on JazzCash.

#### 3. Backup and recovery — the 3-2-1 rule

A **backup** is a copy of data in a safe location (cloud or external drive) so you can **restore** it after hardware damage, ransomware or accidental deletion.

The **3-2-1 backup rule** is a practical standard:

- **3** copies of the data (the original plus two backups).
- **2** different types of storage media (for example laptop disk + USB, or disk + cloud).
- **1** copy **off-site** (a different building or the cloud), so a fire or theft at home does not destroy every copy.

#### 4. Firewall and antivirus software

A **firewall** is a security **barrier** that **monitors network traffic** and **blocks suspicious internet access**. Think of it as a checkpoint on the road between your device (or college lab) and the internet.

**Antivirus** software detects and removes harmful software (viruses, spyware and other malware) from the **local machine**.

Firewall ≈ guards the **door to the network**. Antivirus ≈ searches **inside the house** for intruders already present.

#### WORKED EXAMPLE

**Worked example A — Caesar cipher, shift = 3.** Encrypt the plaintext **SINDH**, then decrypt the result.

Write the alphabet and count three letters forward. Wrap after Z back to A.

| Plain | S | I | N | D | H |
|-------|---|---|---|---|---|
| +3    | V | L | Q | G | K |

Ciphertext = **VLQ GK**.

**Decrypt:** shift each letter **back** by 3: V → S, L → I, Q → N, G → D, K → H. You recover **SINDH**.

Spaces and punctuation are often left unchanged. The same key (3) is used both ways, so Caesar is **symmetric**.

Second check: **CAT** + 3 → C→F, A→D, T→W → **FDW**.

```
def caesar(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr((ord(ch) - base + shift) % 26 + base)
        else:
            result += ch
    return result

print(caesar("SINDH", 3))
print(caesar("VLQ GK", -3))
print(caesar("CAT", 3))
```

#### OUTPUT

```
VLQ GK
SINDH
FDW
```

**Worked example B — 3-2-1 for Fatima's Class XII project in Hyderabad.**

1. **Copy 1 (original):** project folder on her laptop hard disk.
2. **Copy 2 (different media):** the same folder on a USB drive she keeps in her bag.
3. **Copy 3 (off-site):** uploaded to Google Drive (cloud servers are not in her house).

That is 3 copies, 2 media types (internal disk + USB, with cloud as the off-site copy), 1 off-site. When her brother deletes the laptop folder, she restores from USB or Drive.

**Password check:** Zk@2026#Pass has uppercase Z, lowercase letters, digits 2026, and symbols @ and #. fatima has none of those extras — reject it.

3 copies → 2 media types → 1 copy off-site

#### REMEMBER IT

**Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search.** Caesar: shift letters; same key both ways (symmetric).

#### COMMON MISTAKE

**Mistake 1:** Shifting the wrong way, or forgetting wrap-around ( $X, Y, Z + 3 \rightarrow A, B, C$ ).

**Mistake 2:** Two copies on the same laptop (Documents and Desktop) and calling it 3-2-1. That is still one disk in one house.

**Fix:** Count letters on a written alphabet. For backups, different media and one copy that would survive a house fire.

#### اردو میں سمجھیں

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔ انکرپشن پڑھے جانے والے متن کو سائفر ٹیکسٹ بنا دیتی ہے؛ سیزر سائفر میں ہر حرف کو مقررہ جگہوں سے آگے سرکایا جاتا ہے۔ مضبوط پاس ورڈ میں بڑے، چھوٹے حروف، عدد اور علامات ہوں۔ 1-2-3 بیک اپ: تین کاپیاں، دو قسم کے میڈیا، ایک کاپی باہر (آف سائٹ)۔ فائر وال نیٹ ورک ٹریفک روکتی ہے؛ اینٹی وائرس مقامی نقصان دہ سافٹ ویئر ہٹاتا ہے۔

#### CHECK YOURSELF

1. Encrypt PAK with a Caesar shift of 3.
2. What do the 3, the 2 and the 1 stand for in 3-2-1 backup?
3. In asymmetric encryption, which key encrypts and which key decrypts?

Answers: 1. SDN ( $P \rightarrow S, A \rightarrow D, K \rightarrow N$ ). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.

Next lecture: 5.4.1 Common Security Threats (Malware, Phishing, DoS)